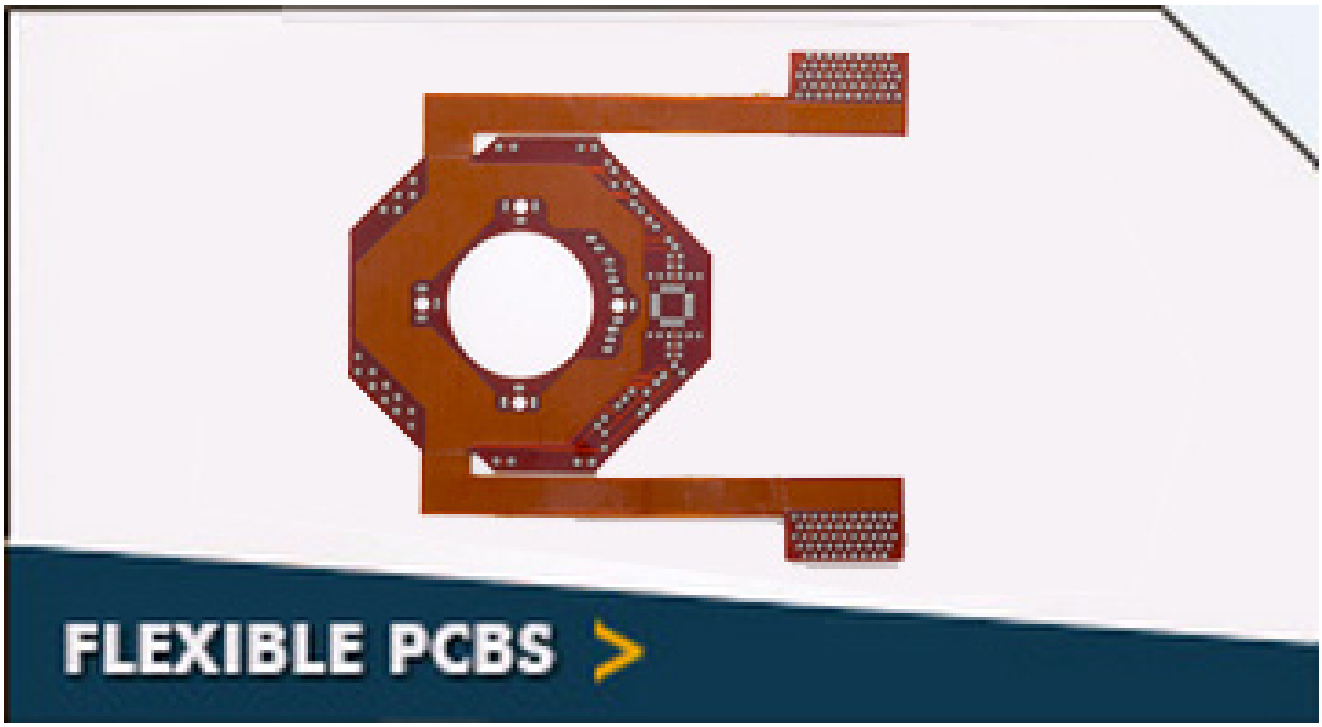
RIGIFLEX TECHNOLOGY, INC.

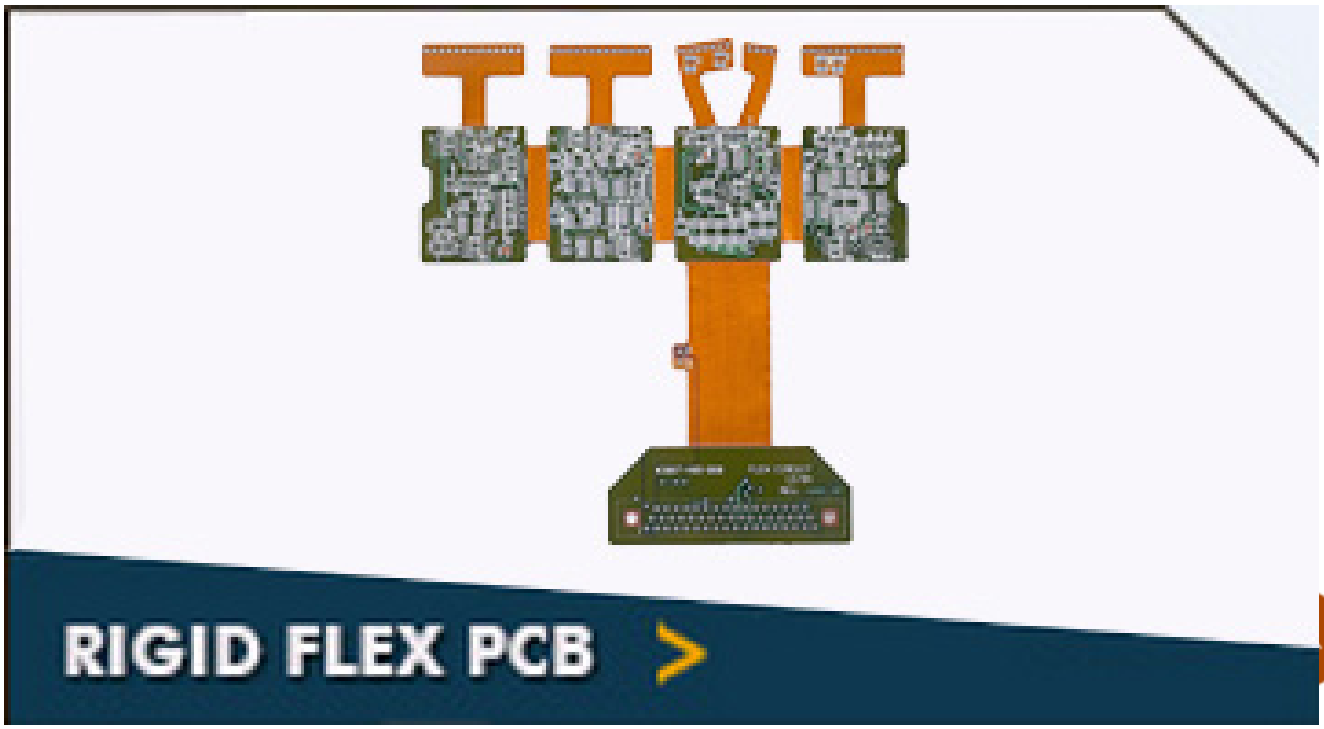
Manufacturers of Rigid, Flex and Rigid-Flex Circuit Boards

(714) 688-1500 (tel:7146881500) pcboards@rigiflex.com (mailto:pcboards@rigiflex.com)

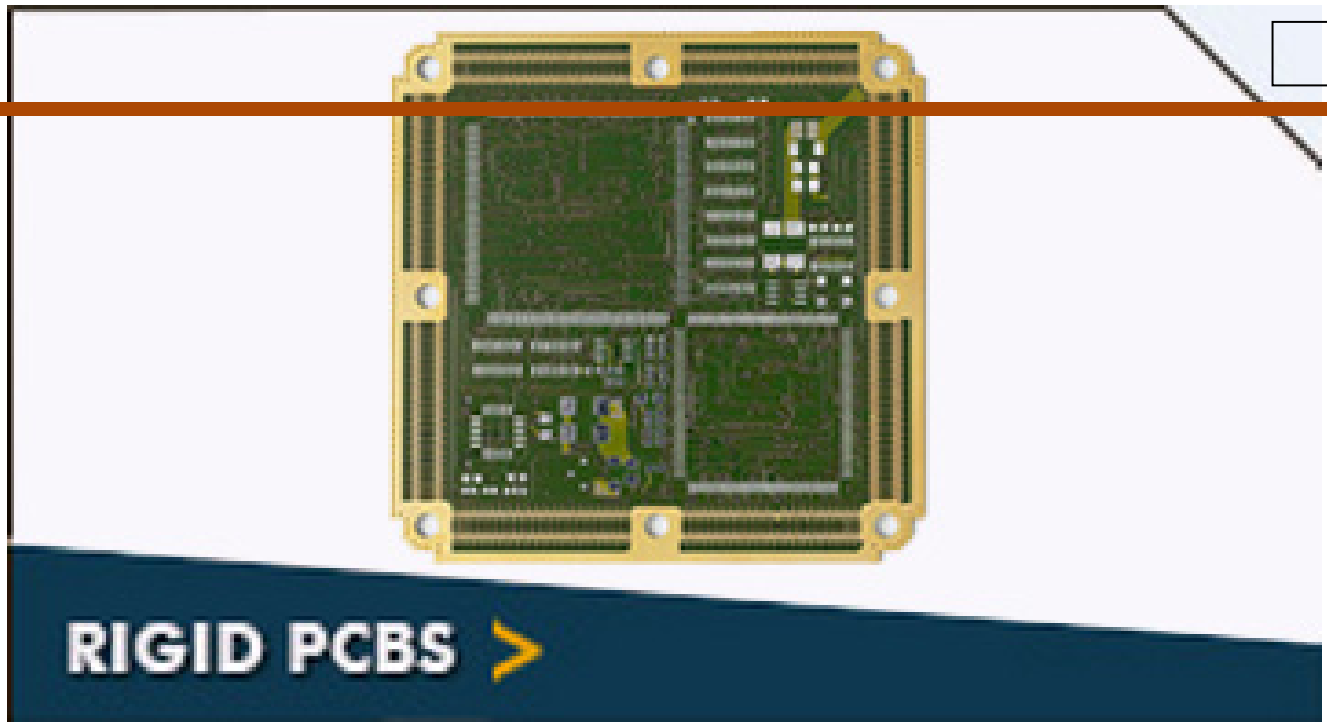
Flex & Rigid-Flex PCBs.. 1-5 Day Quick Turn



(https://www.rigiflex.com/products-services/flexible-pcb/)



(https://www.rigiflex.com/products-services/rigid-flex-pcb/)



(https://www.rigiflex.com/products-services/rigid-pcb/)

Superior & Trusted Flexible, Rigid-Flex & Rigid PCBs for Over 25 years

14 Layer Rigid-Flex PCB



(https://www.rigiflex.com/wp-content/gallery/14-layer-rigid-flex/portfolio_group_Rigiflex-135_original_1.jpg?t=1423031448)

Rigiflex is one of the leading manufacturers of 14 layer rigid-flex PCBs. These PCBs are used for applications, which demand extreme precision and tight tolerances. Over the years, we have helped clients by designing and providing compact sized 14 layer rigid-flex PCBs. These circuit boards comprise flexible inner layers, which are attached using a prepeg film, which resembles to a multilayer flexible circuit. These rigid-flex PCBs are equipped with a board internally, externally, or both – as needed by the circuit.

The rigid components of these circuit boards are utilized for high density device boards. Additionally, flexible circuits allow narrow lines, thereby supporting high density device population. The combination of rigid boards and flexible circuits allow us to offer streamline design, thereby reducing package weight and size.

Typically, these circuit boards are made from FR4 material. We can also provide them in exotic materials such as Rogers Ultralam 2000, Rogers RO4350, Arlon AR350, Nelco 4013, as well as several metal core materials. We use various types of stiffeners to bond flexible layers. They include stainless steel, FR4,

polyimide, etc. We can provide 14 layer rigid-flex PCBs in minimum thickness 0.054" and maximum thickness 0.250". The surface treatment is ENIG, whereas we also provide surface finishes like HASL – leaded and lead free solders, electroless soft gold, electrolytic nickel, immersion silver, carbon ink, etc. We use advanced laser machines, which enable us to deliver accuracy of mil traces, lines, and patterns. The 14 layer rigid-flex PCBs are used in applications such as telecommunication, computers, medical devices, and automobiles.

14 Layer Rigid-Flex	
Specifications	**Technology**
Board Materials	FR4 (Tg – 135C, 145C, 170C) Rogers Ultralam 2000 Rogers RO4350 Rogers RO4003 Polyimide Teflon Black FR4 Arlon AR350 Getek Copper Clad Thermal Substrates Hybrid (Rogers and FR4) BT Epoxy Nelco 4013 Metal Core Materials
Stiffeners	Thermo Set and PSA Based Aluminum FR4 Stainless Steel Polymide
Final PCB Thickness	2 Layer – Min .005" Max .250" 4 Layer – Min .015" Max .250" 6 Layer – Min .025" Max .250" 8 Layer – Min .031" Max .250" 10 Layer – Min .040" Max .250" 12 Layer – Min .047" Max .250" **14 Layer – Min .054" Max .250"** 16 Layer – Min .062" Max .250" 18 Layer – Min .093" Max .250" 20 Layer – Min .125" Max .250" 22 Layer – Min .125" Max .250" >24 Layer – Min .125" Max .250"
Core Thickness	Min .0025"
Maximum PCB Size	2 Layer 20" x 28" Mulitlayer 16" x 26"
Minimum Conductor Space	0.003"
Minimum Conductor Width	0.003"

Minimum Drill Hole Size	0.006″
Finish Plating / Surface Finishes	HASL – Leaded Solder Tin/Nickel HASL – Lead Free Solder Electroless Soft Gold Wire Bondable Soft Gold Nickel Flash Gold Electroless Nickel Immersion Gold OSP Electrolytic Nickel /Hard Gold and Selective Gold Immersion Silver Immersion Tin Carbon Ink ENIG
Finished Copper – Outer Layers	1oz Cu – Min .004″ Trace/Space 2oz Cu – Min .005″ Trace Space 3oz Cu – Min .008″ Trace/Space 4oz Cu – Min .010″ Trace/Space 5oz Cu – Min .012″ Trace/Space
Finished Copper – Inner Layers	.5oz Cu – Min .004″ Trace/Space 1oz Cu – Min .005″ Trace/Space 2oz Cu – Min .006″ Trace/Space 3oz Cu – Min .010″ Trace/Space 4oz Cu – Min .012″ Trace/Space
Inner Layer Clearances	Min .008″ Minimum Finished Hole Size Final Thickness 062″ – .006″ Hole Final Thickness .150″ – .014″ Hole Final Thickness .093″ – .010″ Hole Final Thickness .200″ – .018″ Hole Final Thickness .125″ – .012″ Hole Final Thickness .250″ – .020″ Hole
Gold Fingers	Per IPC-SM-840 LPI Soldermask Peelable Soldermask
Solder Mask Colors	Green/Green Matte White Black/Black Matte Clear Blue Top and Bottom Mix Red One or Both Sides Mix
Silkscreen Type	Thermal Cure Epoxy Ink LPI Ink

Silkscreen Colors	White
	Black
	Yellow Top and Bottom Mix
	Red One or Both Sides Mix
	Blue
CNC Functions	Scoring Edge to Edge Plated Counter bores
	Skip Scoring – .250" Spacing Milling
	30 or 60 Degree Score Angle Blind and Buried Vias
	30 to 100 Degree Countersink Controlled Z Axis Route
	15 to 45 Degree Gold Finger Bevel Castellated Barrels
	Counterbores Offset or Recessed Beveling
	Plated Countersinks
Other PCB Services	Blind and Buried Vias
	Plated Slots Specified Dielectric
	Tented Vias Controlled Impedance
	Solder mask Plugged Vias Via Caps (Solder Mask)
	Conductive Filled Vias
Quality / Testing	Inspect to IPC Class III Continuity Resistance – 10 to 20 Ohms
	Net List Test per IPC-356D Isolation Resistance – 2 to 30 Megaohms
	Test Voltage – 100 to 250 Volts Minimum SMT Pitch 0.5 mm
Tolerances	PTH Hole Size – +/- .002"
	Front to Back – +/- .002"
	NPTH Hole Size – +/- .001"
	Solder Mask – +/- .002"
	Tooling Holes – +/- .001"
	Hole to Pad – +/- .005"

Quick Contact

All Fields are Mandatory (*)

Full Name

Email

Telephone

Type Your Message

aac198

Captcha

Submit



Need it quick? That is our speciality!

FOOTER NAVIGATION

- » Home (https://rigiflex.com/)
- » Company Profile (https://www.rigiflex.com/company-profile)
- » Customers (https://www.rigiflex.com/company-profile/customers)
- » Capabilities (https://www.rigiflex.com/capabilities)
- » Product & Services (https://www.rigiflex.com/products-services)
- » Flexible PCB`s (https://www.rigiflex.com/products-services/flexible-pcb)
- » Rigid-Flex PCB`s (https://www.rigiflex.com/products-services/rigid-flex-pcb)
- » Rigid PCB`s (https://www.rigiflex.com/products-services/rigid-pcb)
- » Quality & Delivery (https://www.rigiflex.com/quality-delivery)
- » Industries Served (https://www.rigiflex.com/industry-served)
- » Contact (https://www.rigiflex.com/contact-us)
- » Privacy Policy (https://www.rigiflex.com/privacy-policy)
- » PCB Quote (https://www.rigiflex.com/pcb-quote)
- » FAQ (https://www.rigiflex.com/faq)
- » Blog (https://rigiflex.com/blog/)
- » Sitemap (https://www.rigiflex.com/sitemap)
- » Case Study (https://www.rigiflex.com/case-study)
- » Certifications (https://www.rigiflex.com/certifications

- XML (https://www.rigiflex.com/sitemap.xml)

FOLLOW US ON

Facebook (https://www.facebook.com/RigiflexTechnologyInc)

Twitter (https://twitter.com/rigiflextech)

1166 N. Grove Street Anaheim, CA 92806

(714) 688-1500 (tel:7146881500)

(714) 688-1505

pcboards@rigiflex.com (mailto:pcboards@rigiflex.com)

© 2025 RIGIFLEX TECHNOLOGY INC. - EXPERTS IN FLEX AND RIGID-FLEX CIRCUIT BOARDS MANUFACTURING